GATE ALL BRANCHES



General Aptitude

QUANTITATIVE APTITUDE

Lecture No.- 15



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Recap of Previous Lecture



Topic Trains, Boats

Topics to be Covered



Topic-1 Races

Topic-2 Data Interpretation

[MCQ]

#Q. If a train travelling at 40 km/hr crosses another train of length 100 meter travelling at 14 km/hr in opposite directions in 30 seconds, then find the length of the train.

Distance = $l_{T_1} + l_{T_2}$

$= (l_{T_1} + 100)$ m

Relative speed $= 54 \text{ km/hr} \times \frac{5}{18} = 15 \text{ m/sec}$

Time = 30 seconds

$D = S \times T \Rightarrow l_{T_1} + l_{T_2} = 15 \times 30$

$l_{T_1} + 100 = 450$

$l_{T_1} = 450 - 100$

$= 350$ m

Assignment

[MCQ]

$P = 520 - 280$

$= 240\text{ m}$

#Q. A train running at 52 km/hr takes 36 seconds to pass a platform. Next it takes 24 seconds to cross a man walking at the platform with 10 km/hr in the same direction. Find the length of the platform.

Assignment

Train Crossing Platform

$D = l_T + l_P$

$= 280 + l_P$

$S = 52\text{ km/hr}$

$T = 36\text{ sec}$

$280 + l_P = 52 \times \frac{5}{18} \times 36$

$280 + l_P = 520$

Train Crossing Man

$D = l \text{ of Train}$

$R.S. = 42\text{ km/hr}$

$T = 24$

$l \text{ of Train} = 42 \times \frac{5}{18} \times 24$

$= 280\text{ m}$

[MCQ]

#Q. A man rows his boat downstream @ 18 km/hr & upstream @ 10 km/hr.

Find the speed of boat in still water.

Assignment

$$S.B. = \frac{D.S + U.S}{2}$$

$$= \frac{18 + 10}{2}$$

$$= \frac{28}{2}$$

$$= 14 \text{ km/hr}$$

A. 12 km/hr

B. 10 km/hr

C. 14 km/hr

D. 16 km/hr



[MCQ]

#Q. A boat traveled from A to B and back to A from B in 5 hours. If the speed of boat in still water and the speed of stream be 7.5 kmph and 1.5 kmph, then what is the distance between A and B?

Assignment

'x' km

A. 18 km

B. 10 km

C. 16 km

D. 5 km

$$\frac{x}{9} + \frac{x}{6} = 5$$

$$\Rightarrow \frac{2x + 3x}{18} = 5$$

$$\Rightarrow \frac{5x}{18} = 5 \quad \therefore x = 18$$

Races

$\frac{A}{B} \times \frac{B}{C} = \frac{1000}{900} \times \frac{1000}{950}$

$\Rightarrow \frac{A}{C} = \frac{1000}{855}$

1000 − 855

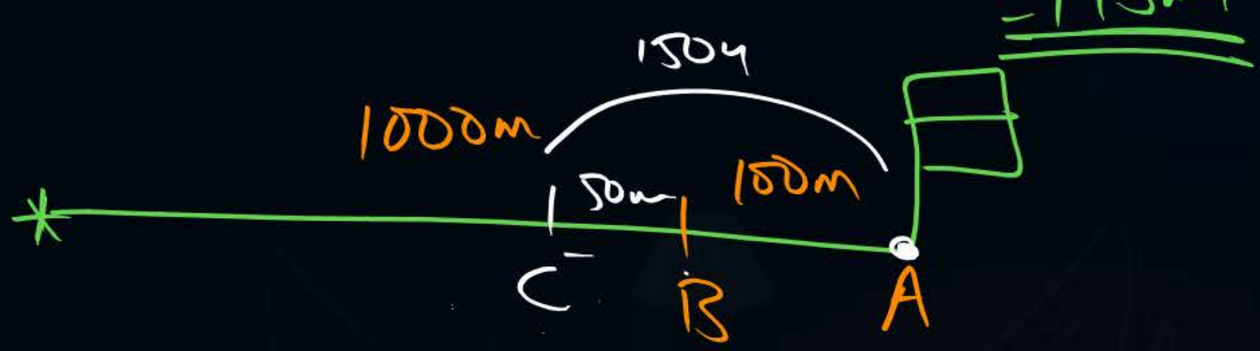
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#Q. In a km race A defeats B by 100 meters and B defeats C by 50 meters, then A defeats C by how many meters?

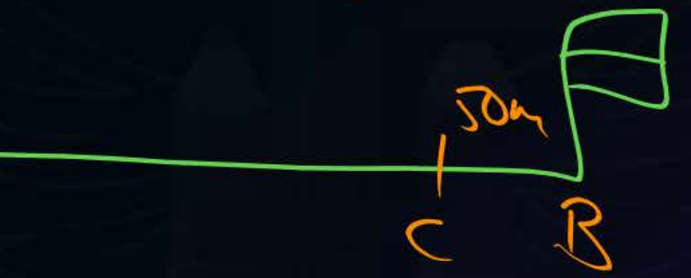
$\frac{A}{B} = \frac{1000}{900}$ w

$\frac{B}{C} = \frac{1000}{950}$ w

= 145m







[MCQ]

#Q. In a km race A defeats (beats) B by 200 meters and B defeats C by 100 meters, then A defeats C by how many meters?

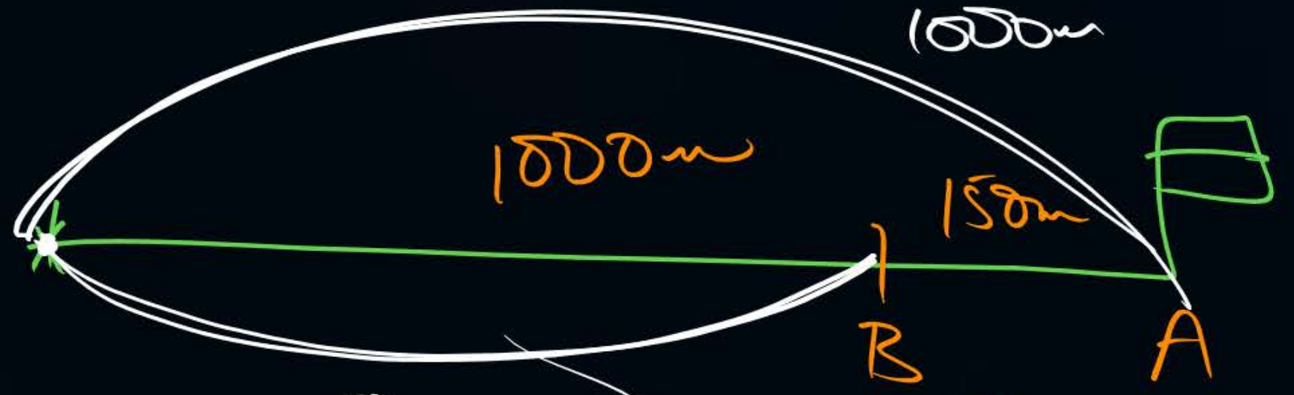
$$\frac{A}{C} = \frac{1000}{800} \times \frac{1000}{900}$$

$$\frac{A}{C} = \frac{1000}{720}$$

280m

[MCQ]

#Q. In a km race A defeats B by 150 meters or 3seconds. Find the speed of A.



$\frac{850}{50}$

$= 17$ seconds

$S_B = \frac{150}{3} = 50$ m/sec (3 sec)

$S_A = \frac{1000}{17} = 58.82$ m/sec

[MCQ]

#Q. In a km race A defeats B by 200 meters or 5seconds whereas B defeats C by 100m. Find the speed of C.

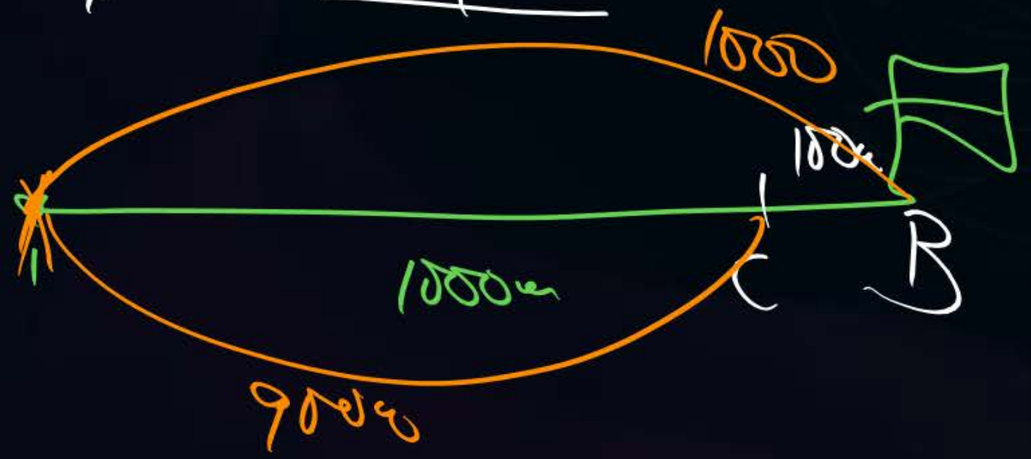


$\frac{1000}{40} = 25$ sec

$S_B = \frac{1000}{5} = 40$ m/sec

$S_C = \frac{900}{25}$

$= 36$ sec



Data Interpretation

Graph

Types:

The numerical data pertaining to any event can be presented by any one or more of the following methods.

1) Tables
2) Line Graphs
3) Bar Graphs or Bar Charts
4) Pie Charts or Circle Graphs

Line Graph:

Study the following graph carefully and answer the questions:



Marks scored by a student out of 500 in the Unit test held at different months of academic year 2019

[MCQ]

#Q. What is the percentage of marks scored by the student in the unit test of August and October taken together?

$\frac{755}{1000} \times 100$

75.5%

A 50%

B 83.3%

C 60%

D 75.5%



Marks scored by a student out of 500 in the Unit test held at different months of academic year 2019

[MCQ]

#Q. What are the average of marks obtained by the student in the unit test of given academic year?

$A = \frac{\text{Sum}}{\text{No.}}$

$= \frac{2285}{6}$

$= 380.833$

A 361.6

B 380.8

C 377.5

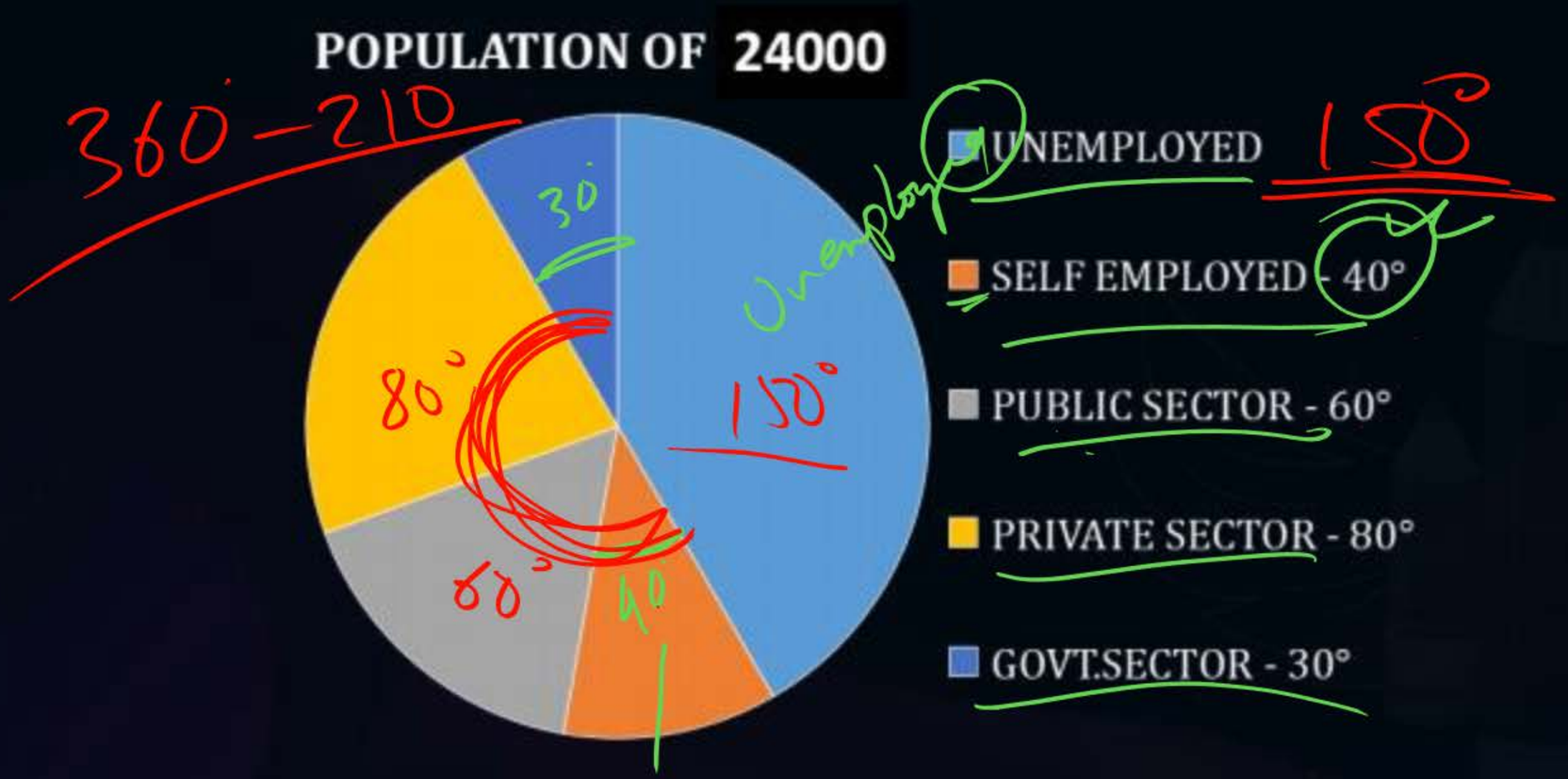
D 400.0



Marks scored by a student out of 500 in the Unit test held at different months of academic year 2019

Pie Chart:

Study the following graph carefully and answer the questions:



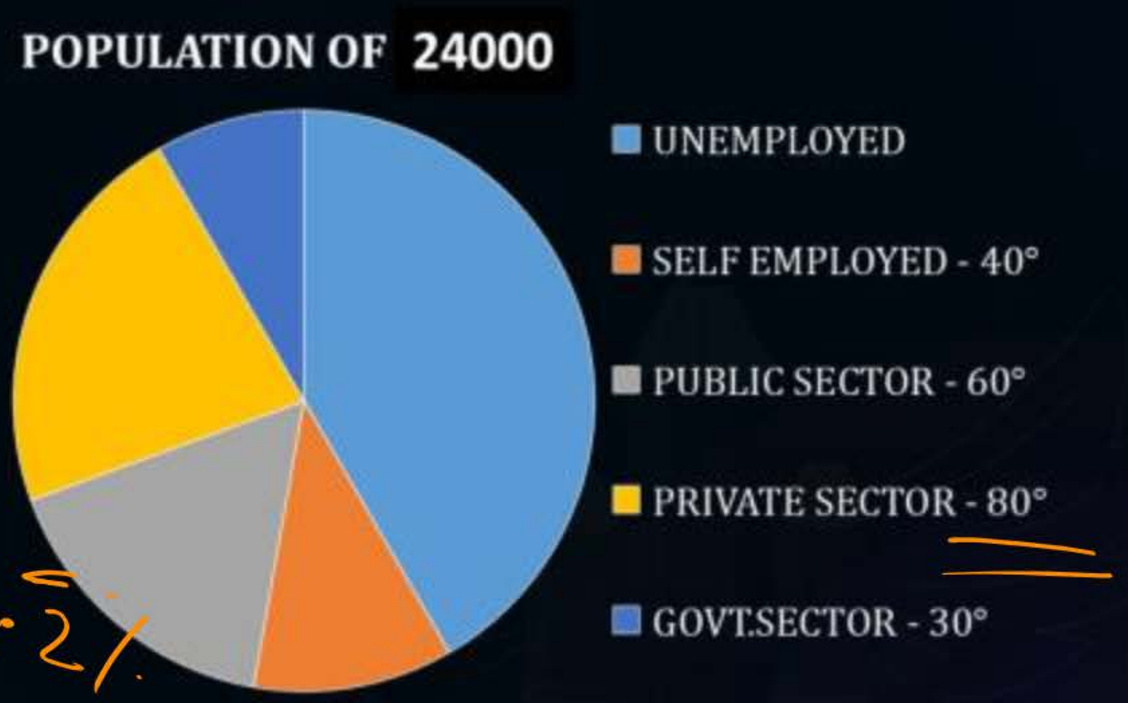
[MCQ]

#Q. If private sector employees are to be mentioned in above chart in the form of percentage, it would be .....%.

below

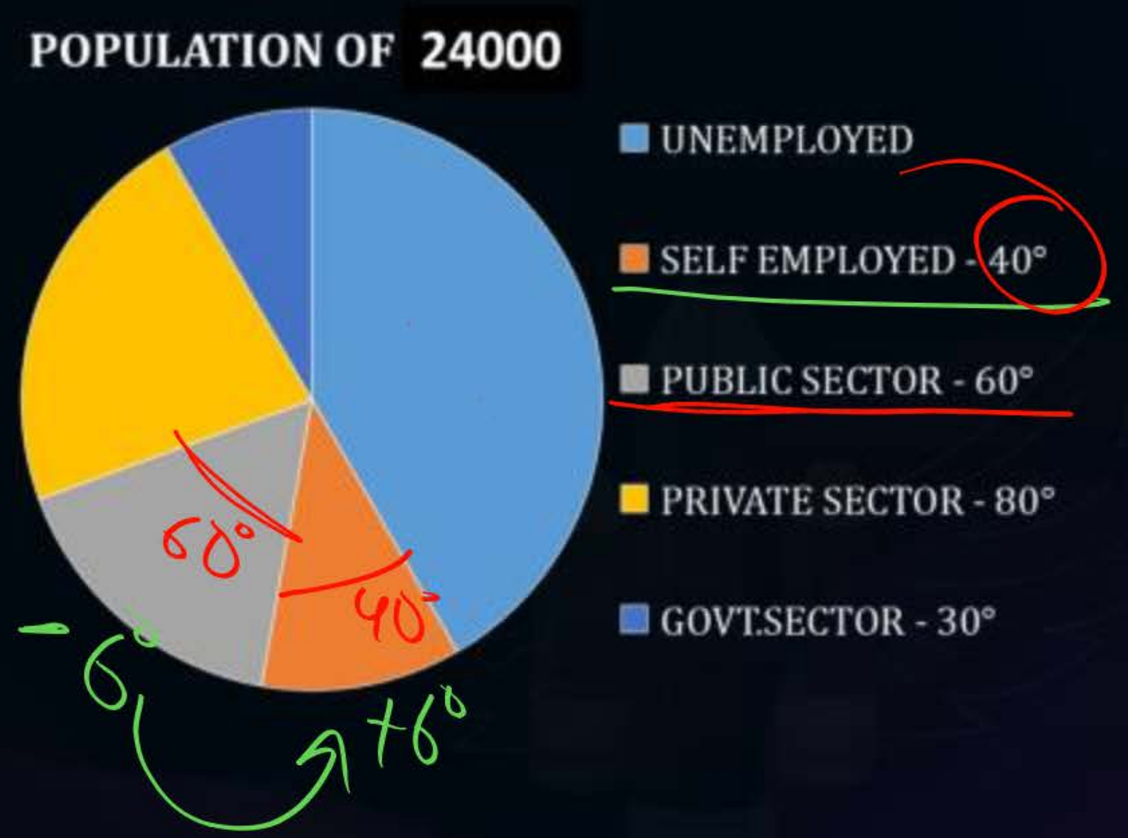
$$\frac{80}{360} \times 100$$

$$= \frac{200}{9} = 22.\overline{2}\%$$



[MCQ]

#Q. If 10% of public sector people resign and becomes self-employed, find the increase percent in self-employed population.



[MCQ]

#Q. What percentage of employed people are working in government sector?

$$\frac{30^\circ}{240^\circ} \times 100$$

$$= \frac{100}{7} = 14.285$$



[MCQ]

#Q. Find the number of people who are unemployed.



Tabular form:

Study the following table and answer the questions:

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

[MCQ]

#Q. Who among the following employees earns maximum bonus in comparison to his total income?

Assignment

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

A P

B Q

C R

D S

[MCQ]

#Q. The income from overtime is what percent of income from arrears in the case of employee P?

Assignment

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

A 90

B 80

C 40

D 75

[MCQ]

#Q. How many employee have their salary more than 3 times the income from bonus?

Assignment

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

A One

B Five

C Three

D None

[MCQ]

#Q. Who among the following employees has maximum percentage of his salary out of the total income?

Assignment

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

A T

B P

C S

D R

[MCQ]

#Q. Who among the following employees has minimum ratio of income from arrears to the income from salary?

Assignment

Source of Income	Employees				
	P	Q	R	S	T
Salary	400	200	700	300	400
Bonus	80	40	150	80	100
Overtime	180	70	200	170	200
Arrears	200	180	400	140	250
Misc	40	10	50	10	50
TOTAL	900	500	1500	700	1000

A P

B Q

C T

D S



2 mins Summary

Topic Races & Data Interpretation

THANK - YOU